1. Sort an array in ascending and descending order.
   Code:
```
package Day3;
import java.util.Arrays;

public class SortArray {
    public static void main(String[] args) {
        int[] arr = {5, 2, 9, 1, 5, 6};

        // Ascending
        Arrays.sort(arr);
        System.out.println("Ascending: " + Arrays.toString(arr));

        // Descending
        int[] desc = arr.clone();
        for (int i = 0; i < desc.length / 2; i++) {
            int temp = desc[i];
            desc[i] = desc[desc.length - 1 - i];
            desc[desc.length - 1 - i] = temp;
        }
        System.out.println("Descending: " + Arrays.toString(desc));
    }
}
```
2. Remove duplicate elements from an array.
   Code:
```
package Day3;
import java.util.LinkedHashSet;

public class RemoveDuplicates {
    public static void main(String[] args) {
        int[] arr = {1, 2, 2, 3, 4, 4, 5};
        LinkedHashSet<Integer> set = new LinkedHashSet<>();
        for (int num : arr) {
            set.add(num);
        }
        System.out.println("Without duplicates: " + set);
    }
}
```
3. Find largest element in each row.
   Code:
```
package Day3;

public class LargestInRow {
    public static void main(String[] args) {
        int[][] matrix = {
            {3, 5, 7},
            {1, 9, 2},
            {8, 6, 4}
        };

        for (int i = 0; i < matrix.length; i++) {
            int max = matrix[i][0];
            for (int j = 1; j < matrix[i].length; j++) {
```

```
            if (matrix[i][j] > max) {
                max = matrix[i][j];
            }
        }
        System.out.println("Largest in row " + (i+1) + ": " + max);
    }
  }
}
```

4. Check if array is palindrome.

   Code:

```
package Day3;

public class PalindromeArray {
  public static void main(String[] args) {
    int[] arr = {1, 2, 3, 2, 1};
    boolean isPalindrome = true;

    for (int i = 0; i < arr.length / 2; i++) {
      if (arr[i] != arr[arr.length - 1 - i]) {
        isPalindrome = false;
        break;
      }
    }

    System.out.println("Is palindrome? " + isPalindrome);
  }
}
```

5. Rotate matrix by 90 degrees.

   Code:

```
package Day3;

public class RotateMatrix {
  public static void main(String[] args) {
    int[][] matrix = {
      {1, 2, 3},
      {4, 5, 6},
      {7, 8, 9}
    };

    int n = matrix.length;

    // Transpose
    for (int i = 0; i < n; i++) {
      for (int j = i; j < n; j++) {
        int temp = matrix[i][j];
        matrix[i][j] = matrix[j][i];
        matrix[j][i] = temp;
      }
    }

    // Reverse each row
    for (int i = 0; i < n; i++) {
      for (int j = 0; j < n / 2; j++) {
```

```
            int temp = matrix[i][j];
            matrix[i][j] = matrix[i][n - 1 - j];
            matrix[i][n - 1 - j] = temp;
        }
    }

    // Print rotated matrix
    System.out.println("Rotated 90°:");
    for (int[] row : matrix) {
        for (int val : row) {
            System.out.print(val + " ");
        }
        System.out.println();
    }
  }
}
```